

Tomo

Read, write, and connect.

Background

Tomo serves a large global community of fanfiction readers and writers who are highly active but lack a centralized modern platform. Most rely on a small number of existing platforms that are often outdated, poorly optimized for mobile, or no longer actively maintained. As a result, the community feels “homeless” despite its size and engagement. These users prioritize creative freedom and seek a platform designed specifically for fanfiction that supports long-form reading, detailed tagging systems, and active community discussions.

Problem Statement

Although the fanfiction community continues to grow, the digital platforms supporting it have not kept pace. Users face several key challenges including limited platform choices, outdated or abandoned apps, and poor mobile experiences due to a lack of app-first design. Features that modern users expect such as offline reading, real-time interaction, and seamless syncing are often missing. In addition, the social aspect of fandom is fragmented with discussions happening on separate platforms, making it harder for readers and writers to connect.

Solution

Tomo addresses these gaps by providing a modern all-in-one mobile application where users can read, write, and interact within a single high-performance ecosystem tailored to the needs of the fanfiction community.

Module 1: Client Tier (Local Vault)						
Test Case ID		Tomo-CL				
Module Name		Client Tier				
Test Scenario / Description		handles user interaction and stores data locally for offline access using an encrypted database.				
Test Scenario / Description		Save story for offline reading				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-CL-01	Open a story		S001-the complication	Story accessible offline	Story successfully opened without internet	Pass
	Tap "Download"					
	Disable internet					
	Open Library → Download					
Test Scenario / Description		resume reading progress offline				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-CL-02	Open a story		S001- Chapter 1	Story accessible offline	App resumed at correct scroll position	Pass
	Scroll to mid-point					
	Close app					
	Reopen without internet					
Test Scenario / Description		Auto-cache next chapter				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-CL-03	Open Chapter 1		S001-the complication	Next chapter loads without internet	Next chapter loaded, minor delay observed	Pass
	Wait for preload					
	Disable internet					
	Navigate to next chapter					
Module 2: Sync Service						
Test Case ID		Tomo-SYNC				
Module Name		Sync Service				
Test Scenario / Description		Synchronizes local and cloud data while resolving conflicts between devices.				
Test Scenario / Description		Sync reading progress after reconnect				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SYNC-01	Read story offline		S001: 40%	Progress updates in cloud database	Progress updated after reconnect	Pass
	Change progress					
	Reconnect internet					
Test Scenario / Description		Resolve conflict from multiple devices				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SYNC-02	Read on Device A (offline)		Conflicting progress values	Conflict resolved (latest version applied)	Latest progress applied (simulated)	Pass
	Read on Device B (online)					
	Reconnect Device A					
Test Scenario / Description		Sync downloaded story to cloud				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SYNC-03	Download story offline		S001-the complication	Story syncs with cloud storage	Sync triggered successfully (simulated)	Pass
	Reconnect internet					
	Disable internet					
	Open Library → Download					

Module 3: Story Service						
Test Case ID		Tomo-STORY				
Module Name		Story Service				
Test Scenario / Description		Manages story retrieval, rendering, and search across the archive.				
Test Scenario / Description		Search with include/exclude tags				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-STORY-01	Open search		Tags	Filtered stories returned correctly	Results matched include/exclude filters	Pass
	Include tag "Romance"					
	Exclude tag "Angst"					
Test Scenario / Description		Load and render story content				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-STORY-02	Select story		S001	Content renders correctly (Markdown supported)	Markdown rendered correctly	Pass
	Open chapter					
	Reconnect internet					
Test Scenario / Description		Perform search on large dataset				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-STORY-03	Input keywords in search bar		Large archive dataset	Results returned quickly (within acceptable time)	Slight delay under large dataset	Pass
	Enter search query					
Module 4: Social Service						
Test Case ID		Tomo-SOC				
Module Name		Social Service				
Test Scenario / Description		Handles community features such as chat, forums, and moderation.				
Test Scenario / Description		Send message in Circle				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SOC-01	Open Circle chat		"Hello"	Message appears in real-time	Message delivered instantly	Pass
	Send message					
Test Scenario / Description		Block user interaction				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SOC-02	Block User B		User B-ID	Interaction is prevented	Block successfully prevented messaging	Pass
	Attempt communication					
Test Scenario / Description		Detect harmful content (Fuzzy Filter)				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-SOC-03	Send altered offensive message		B#@d w0rd	Harmful content filtered	Offensive text flagged (partial detection)	Partial Pass

MODULE 5: INTELLIGENCE SERVICE						
Test Case ID		Tomo-INT				
Module Name		Intelligence Service				
Test Scenario / Description		Generates personalized recommendations using user behavior and Library DNA.				
Test Scenario / Description		Generate personalized feed				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-INT-01	Perform reading activities		User history	Personalized recommendations appear	Recommendations matched user activity (simulated)	Pass
	Refresh home page					
Test Scenario / Description		Apply PII masking				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-INT-02	Send user data to service		Name, Email	Sensitive data is removed before processing	Sensitive data excluded (simulated)	Pass
Test Scenario / Description		Apply diversity in recommendations				
Test Case ID	Test Steps		Test Data	Expected Result	Actual Result	Status
Tomo-INT-03	Repeatedly read same genre		Genre = Angst	Diverse recommendations appear	Some variation observed, limited diversity	Partial Pass

Bug Analysis and Reflection	
A. Delay in Loading Large Data (Story Service)	B. Inconsistent Detection of Harmful Content (Fuzzy Filter)
Issue: performing A slight delay was observed when loading stories or performing searches on large datasets.	Issue: The system failed to consistently detect modified or obfuscated offensive words (e.g., altered spellings).
Possible Cause: ■ Inefficient query handling or indexing ■ Network latency when retrieving data from the backend ■ Lack of optimized caching for frequently accessed content	Possible Cause: ■ The filtering algorithm may rely on simple pattern matching ■ Lack of advanced natural language processing (NLP) techniques ■ Insufficient training data for detecting variations of harmful words
Recommendation: ■ Improve indexing strategies in PostgreSQL (e.g., optimize B-Tree usage) ■ Utilize Redis caching for frequently accessed stories and search results ■ Optimize queries and integrate better pagination or lazy loading	Recommendation: ■ Enhance the Fuzzy Filter using more advanced pattern recognition or NLP techniques ■ Expand the database of offensive word variations ■ Implement machine learning-based moderation for improved accuracy
C. Limited Diversity in Recommendations (Intelligence Service)	
Issue: The recommendation system occasionally produced repetitive or overly similar content.	
Possible Cause: ■ Over-reliance on dominant user preferences in "Library DNA" ■ Lack of diversity injection or randomness in recommendation logic ■ Insufficient weighting of secondary tags or categories	
Recommendation: ■ Introduce a stronger diversity algorithm (e.g., random genre injection or "exploration factor") ■ Adjust weighting to balance user preference and content variety ■ Periodically refresh recommendation pools to avoid repetition	